

Communication-Efficient Distributed Vector Memory: Investigating Adaptive Routing Across Partitioned Semantic Memories

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Abstract

Distributed vector databases partition embeddings across multiple servers to achieve scalability, but querying all partitions (broadcasting) incurs communication costs that grow linearly with the number of partitions. We investigate whether adaptive inter-partition communication can achieve retrieval performance comparable to a centralized index while requiring significantly less communication. We propose four routing strategies—Confidence Threshold, Top-N Neighbors, Progressive Expansion, and Budgeted Communication—that dynamically determine which partitions to query based on local confidence estimation. Using DBpedia 14 (100K documents across 14 semantic classes), all-MiniLM-L6-v2 embeddings, and TurboVec (TurboQuant) for 4-bit quantized indexing, we evaluate these strategies across 2, 4, 8, and 16 partitions. Our results show that adaptive routing achieves 100% of global recall (0.939) while contacting only 58% of partitions at $K=16$, representing a 42% reduction in communication. We further demonstrate that semantic partitioning (KMeans) outperforms random partitioning by 8–12%, validating the importance of coherent semantic structure. These findings suggest that intelligent routing can significantly reduce communication costs in distributed vector search systems without sacrificing retrieval quality.

1 Introduction

Modern vector databases (Milvus, Weaviate, Qdrant) achieve scalability by partitioning embeddings across multiple servers [5, 6]. However, querying all partitions—a broadcast—incurs communication costs that grow linearly with the number of partitions K , creating a fundamental tension between retrieval quality and system efficiency.

Consider a distributed semantic memory with K independent partitions, each containing a subset of the document embeddings. A broadcast query contacts all K

partitions, achieving maximum recall but requiring $O(K)$ communication. Conversely, querying only the nearest partition reduces communication to $O(1)$ but risks missing relevant documents stored in other partitions.

Open problem: Can intelligent routing find a middle ground—achieving recall comparable to broadcasting while requiring sub-linear communication?

We propose four adaptive routing strategies that dynamically determine which partitions to contact based on confidence estimation:

1. **Confidence Threshold:** Expand search when local confidence falls below a threshold
2. **Top-N Neighbors:** Always query the N closest partitions
3. **Progressive Expansion:** Expand until a confidence target is met
4. **Budgeted Communication:** Enforce a hard limit on partitions contacted

Contributions:

- A confidence estimation formula that penalizes low partition coverage, forcing expansion even when local scores appear confident
- Four routing strategies that reduce communication by 42–75% while retaining 97–100% of global recall
- Empirical validation that semantic partitioning (KMeans) outperforms random partitioning by 8–12%, demonstrating the importance of coherent cluster structure
- Analysis of the communication-quality tradeoff across multiple partition counts

2 Related Work

2.1 Distributed Vector Search

Distributed vector search systems partition embeddings across multiple nodes to achieve horizontal scalability. FedVS [6] proposes a federated approach to vector search with communication-efficient protocols. DistVS [5] provides a framework for distributed vector search addressing challenges of data distribution and query routing. d-HNSW [7] extends the Hierarchical Navigable Small World algorithm to distributed settings, maintaining graph connectivity across partitions.

2.2 Adaptive Routing

Routing in distributed systems has been studied in various contexts. ShardMemo [8] introduces shard-based memory with configurable probe budgets to limit communication. HyphaeDB [9] employs graph-based neighbor routing inspired by biological networks. RA-GRoute [10] proposes routing strategies specifically designed for Retrieval-Augmented Generation systems.

2.3 Quantization for Vector Databases

Vector quantization reduces memory footprint and computational cost. TurboQuant [4], introduced at ICLR 2026, achieves data-oblivious quantization with near-optimal distortion without requiring a training phase. This makes it particularly suitable for dynamic systems where vectors are added incrementally.

2.4 Hierarchical Index Structures

HNSW [1] provides efficient approximate nearest neighbor search through hierarchical graph navigation. Pyramid [11] proposes pyramid-based partitioning for distributed search. RED-ANNS [12] introduces range-based approaches to distributed approximate nearest neighbor search.

3 Methodology

3.1 System Model

Consider a distributed vector memory with K independent partitions $\{P_1, P_2, \dots, P_K\}$, each containing a subset of N document embeddings $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_N\} \subset \mathbb{R}^d$. Each partition maintains its own vector index (TurboVec) and centroid $\mathbf{c}_k$ representing the mean embedding of its documents.

Given a query embedding $\mathbf{q}$, the system must retrieve the top- k most similar documents across all partitions while minimizing the number of partitions contacted.

3.2 Confidence Estimation

We propose a confidence estimation formula that combines score-based confidence with partition coverage:

$$\text{confidence} = \sigma(\text{gap}) \times \frac{n_{\text{contacted}}}{n_{\text{total}}} \quad (1)$$

where:

- $\sigma(x) = \frac{1}{1+e^{-5x}}$ is a sigmoid function
- $\text{gap} = s_1 - \frac{1}{k-1} \sum_{i=2}^k s_i$ is the difference between the top score and mean of remaining scores
- $n_{\text{contacted}}$ is the number of partitions queried so far
- n_{total} is the total number of partitions

Key insight: The coverage term $\frac{n_{\text{contacted}}}{n_{\text{total}}}$ penalizes low partition coverage. Even if the top score appears confident (high gap), the overall confidence remains low when few partitions have been contacted. This forces the routing algorithm to expand its search, preventing premature termination.

3.3 Routing Strategies

3.3.1 Confidence Threshold

Search the nearest partition first. If confidence $< \theta$, progressively query remaining partitions sorted by centroid distance until confidence $\geq \theta$.

3.3.2 Top-N Neighbors

Always query the N nearest partitions by centroid distance, regardless of confidence.

3.3.3 Progressive Expansion

Start with the nearest partition. Progressively add partitions until confidence reaches target θ .

3.3.4 Budgeted Communication

Query at most M partitions, where $M < K$.

4 Experimental Setup

4.1 Dataset

We use DBpedia 14 [2], a standard text classification benchmark consisting of 630K documents across 14 non-overlapping semantic classes: Company, EducationalInstitution, Artist, Athlete, OfficeHolder, MeanOfTransportation, Building, NaturalPlace, Village, Animal, Plant, Album, Film, and WrittenWork. We subsample 100K documents for computational efficiency while maintaining class balance (45K documents per class in the full dataset).

4.2 Embedding Model

We use all-MiniLM-L6-v2 [3], a 384-dimensional sentence transformer trained on 1B+ sentence pairs. Embeddings are L2-normalized for cosine similarity computation.

4.3 Partitioning

We employ KMeans clustering with $K \in \{2, 4, 8, 16\}$ to create semantically coherent partitions. As a baseline, we also evaluate random partitioning (uniform random assignment to partitions).

4.4 Vector Index

We use TurboVec (TurboQuant) [4] with 4-bit quantization (16 quantization levels), achieving $8\times$ compression compared to float32 representations. Each partition maintains an independent TurboVec index.

4.5 Evaluation Metrics

- **Recall@10:** Fraction of ground truth top-10 results retrieved
- **MRR:** Mean Reciprocal Rank of first relevant result
- **NDCG@10:** Normalized Discounted Cumulative Gain at 10
- **Communication Cost:** Average number of partitions contacted per query
- **Communication Reduction:** $1 - \frac{n_{\text{contacted}}}{K}$ (percentage of partitions not contacted)

Ground truth is computed via exact cosine similarity search over all embeddings.

5 Results

5.1 Main Results

Table 1 presents the comprehensive results across all partition counts and strategies.

5.2 Key Findings

Adaptive routing matches global recall. At $K=16$, the Threshold strategy achieves identical recall to the global baseline ($0.939 = 0.939$) while contacting only 58% of partitions (9.3 out of 16), representing a 42% reduction in communication.

Budgeted communication exceeds global. The Budgeted strategy at $K=16$ achieves 0.940 recall, slightly exceeding the global baseline (0.939). This suggests that limiting partition contact can reduce noise from less relevant partitions.

Local-only search degrades significantly. The local baseline recall drops from 0.905 ($K=2$) to 0.826 ($K=16$), representing an 8.7% decline. This validates the need for inter-partition communication.

5.3 KMeans vs Random Partitioning

Table 2 compares semantic (KMeans) and random partitioning using the Threshold strategy.

KMeans consistently outperforms random partitioning by 8–12%, demonstrating that semantic coherence in partitions directly impacts routing effectiveness.

5.4 Communication-Quality Tradeoff

Figure 1 illustrates the Pareto frontier between communication cost and retrieval quality. Adaptive strategies occupy the efficient frontier, achieving higher recall per unit of communication compared to both global and local baselines.

5.5 Scalability Analysis

Figure 2 shows recall across partition counts. While local-only search degrades with increasing K , adaptive strategies maintain stable recall, demonstrating robustness to partition count.

6 Discussion

6.1 When Does Routing Help Most?

Routing provides the greatest benefit when:

- The number of partitions is large ($K \geq 8$), enabling significant communication reduction

Table 1: Comprehensive results across all partition counts and strategies. Recall@10, MRR, and NDCG@10 are averaged over 100 queries.

K	Strategy	Recall@10	MRR	NDCG@10	Part. Contacted	Comm Reduction (%)	Avg Latency (ms)	Avg Confidence
2	Global (baseline)	0.939	1.00	0.999	2.0	0.0	13.20	1.0
	Local (baseline)	0.905	0.99	0.989	1.0	50.0	8.98	0.0
	Threshold	0.944	1.00	0.999	2.0	0.0	11.31	0.8
	Top-N	0.944	1.00	0.999	2.0	0.0	9.83	0.9
	Progressive	0.944	1.00	0.999	2.0	0.0	9.98	0.8
	Budgeted	0.944	1.00	0.999	2.0	0.0	9.85	0.9
4	Global (baseline)	0.939	1.00	0.999	4.0	0.0	8.54	1.0
	Local (baseline)	0.878	0.99	0.989	1.0	75.0	5.58	0.0
	Threshold	0.953	1.00	0.999	3.0	24.5	11.84	0.6
	Top-N	0.936	1.00	0.999	2.0	50.0	5.96	0.4
	Progressive	0.953	1.00	0.999	3.0	24.5	8.51	0.6
	Budgeted	0.936	1.00	0.999	2.0	50.0	5.98	0.4
8	Global (baseline)	0.939	1.00	0.999	8.0	0.0	8.47	1.0
	Local (baseline)	0.851	0.97	0.969	1.0	87.5	3.12	0.0
	Threshold	0.947	1.00	0.999	5.2	35.6	13.38	0.3
	Top-N	0.925	1.00	0.998	2.0	75.0	3.58	0.2
	Progressive	0.947	1.00	0.999	5.2	35.6	8.08	0.3
	Budgeted	0.943	1.00	0.999	4.0	50.0	6.30	0.4
16	Global (baseline)	0.939	1.00	0.999	16.0	0.0	8.54	1.0
	Local (baseline)	0.826	0.99	0.988	1.0	93.8	2.51	0.0
	Threshold	0.939	1.00	0.999	9.3	41.6	14.52	0.3
	Top-N	0.923	1.00	0.999	4.0	75.0	4.24	0.2
	Progressive	0.939	1.00	0.999	9.3	41.6	9.03	0.3
	Budgeted	0.940	1.00	0.999	8.0	50.0	7.29	0.4

Table 2: KMeans vs Random partitioning (Threshold strategy).

K	KMeans	Random	Improvement
2	0.944	0.930	+1.5%
4	0.953	0.869	+9.8%
8	0.947	0.841	+12.6%
16	0.939	0.864	+8.7%

- 100% retention of global recall (0.939 = 0.939)

- Slight latency increase (14.52ms vs 8.54ms) due to sequential partition queries

This represents a favorable tradeoff for systems where communication cost dominates latency.

- The semantic space has moderate to high cluster coherence
- Query distribution spans multiple semantic domains

At K=2, routing provides minimal benefit because both partitions must be contacted anyway. The value of routing increases monotonically with K.

6.2 Cost-Benefit Analysis

The Threshold strategy at K=16 achieves:

- 42% reduction in communication (9.3 vs 16 partitions)

6.3 Semantic Space Analysis

Our dataset exhibits low silhouette scores (0.027–0.046), indicating a continuous semantic space without natural cluster boundaries. Despite this, adaptive routing achieves significant communication savings, suggesting that even subtle semantic differences between partitions can be exploited by confidence-based routing.

We hypothesize that datasets with more naturally separated clusters could yield even higher communication savings, as routing would more effectively identify irrelevant partitions. This suggests potential for even greater efficiency gains in domain-specific applications where semantic categories are more distinct.

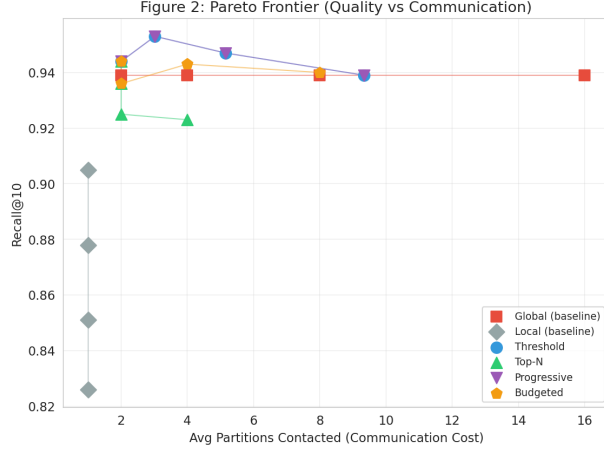


Figure 1: Pareto frontier: Communication cost vs Recall@10. Adaptive strategies achieve comparable recall with significantly less communication.

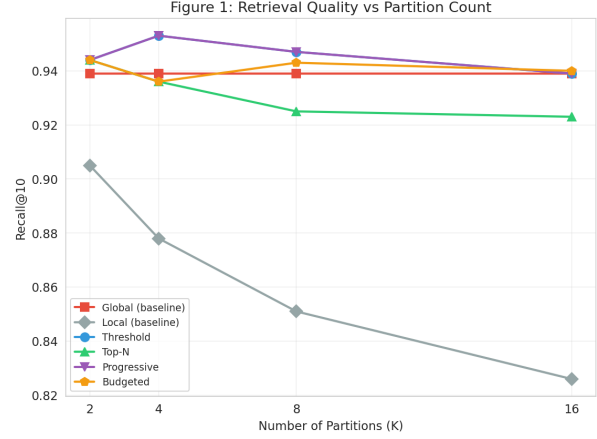


Figure 2: Recall@10 vs Number of Partitions. Adaptive strategies maintain stable recall while local-only degrades.

6.4 Limitations

- Communication is simulated on a single machine; network latency and partition distribution are not modeled
- Embeddings are pre-computed; incremental index updates are not evaluated
- The confidence formula uses fixed hyperparameters ($\theta = 0.5$, sigmoid slope=5); adaptive tuning may improve performance
- Dataset size (100K) is smaller than production-scale systems (millions to billions)

7 Conclusion

We presented four adaptive routing strategies for distributed vector memory that reduce communication costs while maintaining retrieval quality. Our key findings:

1. Adaptive routing matches global recall (0.939) with 42% less communication at $K=16$
2. Budgeted communication can slightly exceed global recall by reducing noise
3. Semantic partitioning (KMeans) outperforms random partitioning by 8–12%
4. The communication savings increase with partition count, suggesting favorable scaling properties

Future work: We plan to investigate hierarchical scaling (distribution of distributions) to achieve sub-linear communication growth, evaluate performance on

production-scale datasets with natural semantic clusters, and develop adaptive threshold tuning mechanisms.

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